

# TB Filter Table

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## Overview

**Turns a table's changing harmonic shape into a moving filter for hollow, glassy, vocal, stepped, and folded tone.**

### What this plug-in solves

A conventional filter offers a familiar cutoff and resonance shape, but complex evolving contours often require many EQ bands and automation lanes. TB Filter Table converts the harmonic profile of a selected single-cycle table frame into a filter response. Moving Frame, choosing a different table, or using the Frame LFO changes the entire contour as one musical gesture, while optional Drive adds a separate nonlinear color after the filter.

### What you can expect

- Continuous or stepped movement through procedural harmonic contours instead of a single static filter slope
- Four distinct phase characters, directional Frame motion, and optional nonlinear Drive color
- A large procedural table library plus drawable Custom tables that can be saved or exchanged separately
- Factory and User presets that recall the complete effect, including the Custom table and LFO state

## How it works

A selected table contains a series of harmonic shapes. Frame chooses a position in that series, and the plug-in turns the chosen shape into a filter contour rather than playing it as an oscillator.

Cutoff places the contour in the spectrum, Resonance changes its contrast, and Phase changes its time and phase character. The Frame LFO can travel through the table automatically, while Drive adds a separate input-dependent color after the filter.

## Quick start

- 1** Load Init or the closest Factory preset and choose a Table whose basic contour suits the source.
- 2** Move Frame with the knob or waterfall until the useful hollow, vocal, glassy, folded, or stepped character appears.
- 3** Set Cutoff with the knob or response marker so the contour sits in the useful register; remember that it places harmonic 24 rather than a normal filter corner.
- 4** Adjust Resonance for contour contrast, then set Mix before deciding whether the source needs Drive.
- 5** Compare MINIMUM, LINEAR, ORIGINAL, and RAW on the same passage and keep the phase character that supports the transients and partial blend.
- 6** Add Morph LFO motion with Type, directional Range, and Rate; reposition the base Frame if the live motion stays pinned at a table end.
- 7** Open EDIT only when Drive or Custom-table tools are needed, and distinguish a .tbft table file from a full .tbftpreset User preset.
- 8** Compare with bypass at a matched listening level, keep DAW delay compensation enabled, and leave headroom: there is no user Output control, and the non-RAW modes can add up to +12 dB of internal automatic makeup.

## Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

### Waterfall and response displays

The upper waterfall shows authored table frames and highlights the live frame used by the audio engine. The TABLE FIR RESPONSE graph below it shows the current steady-state filter response before Drive and before the dry/wet Mix. It is not an input spectrum, output meter, or audio FFT. Drag the waterfall to move the base Frame; drag only the vertical marker in the response graph to change Cutoff.

### Table and Frame

Table chooses CUSTOM or one of the procedural Factory tables. The left and right buttons move through the complete table library and wrap at both ends. Frame is the base position within the chosen table. Empty positions between authored keyframes either interpolate smoothly or select the nearest authored frame when the table uses Stepped movement.

## Cutoff, Resonance, and Phase

Cutoff (H24) places the table's twenty-fourth harmonic at the chosen frequency; it is not the usual low-pass cutoff point. For host-automation compatibility, its public coordinate remains 20 Hz to 20 kHz with a 1 kHz skew centre, while the UI, DSP, and schema 5 state use 200 Hz as the effective lower limit. Resonance changes the contrast and depth of the spectral contour. MINIMUM, LINEAR, and ORIGINAL apply internal automatic makeup capped at +12 dB, while RAW bypasses that makeup. MINIMUM favors a causal response, LINEAR uses a symmetric constant-delay response, ORIGINAL retains more of the source frame's phase character, and RAW turns the waveform more directly into a rough filter kernel. Every phase keeps the same reported alignment, but the audible transient and partial-Mix character can differ.

## Morph LFO

TYPE selects OFF, SINE, TRI, SAW, SQR, or RND. RANGE is directional: positive settings travel upward from the base Frame, negative settings travel downward toward it, and the neutral position stops the offset. It is not a symmetric movement around both sides of the base. RATE is free-running and is not synchronized to DAW tempo. The live Frame remains limited to the table ends, so placing the base too close to the travel direction can flatten part of the motion.

## EDIT and TABLE TOOLS

The actual main view and TABLE TOOLS view are shown together above. EDIT opens this secondary workspace, which reveals Drive, a larger Frame rail, preset/history controls, and Custom-table editing. NEW creates a two-keyframe Custom table. CAPTURE copies the currently rendered frame into the same Custom slot. DRAW FRAME lets you draw the current waveform and commits it to Custom when the gesture ends. CLEAR removes the authored frame at the current slot when Custom is selected. TABLE FILE imports or exports only .tbft table data; it is separate from a full plug-in preset.

## FRAME LFO tools

The FRAME LFO tab exposes the same Type, Range, and Rate controls in a larger work area and shows BASE and LIVE positions separately. Opening or closing either tool view does not change the sound. Keyboard navigation can move Frame in small or larger steps, jump to the ends, cycle tables, open EDIT, and clear a Custom keyframe when the relevant view has focus.

## Preset and history controls

The preset browser groups Essentials, Motion, Character, Showcase, and USER. Previous and Next move through Factory and User presets, while Save User Preset, Undo, and Redo manage complete state changes. The Factory set is Init, Gentle Pad Motion, Hollow Drum Focus, Vocal Formant Lift, Subtle Bus Contour, Slow Vowel Bloom, Liquid Reed, Golden Sweep, Phase Weave, Comb Garden, Warm Ladder, Prime Choir, Glass Tide, Dense Fold, Pulse Steps, Raw Barcode, Mobius Kernel, Prism Bloom, Nested Fold, and Broken Speaker.

## Table library and file types

The Factory table library contains Fundamentals, Harmonic Ladders, Spectral Contours, Geometry, Polynomial Orbits, Phase Symmetry, Pulse Architecture, Fold and Warp, Organic Motion, and Experimental Curves. Each group contains multiple procedural tables with different keyframe densities and Smooth or Stepped behavior. User presets use .tbftpreset and are kept in Documents/TB Filter Table/Presets/User; .tbft files contain only a Custom table and its morph behavior.

## Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

| Control             | What it does and when to use it                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TABLE</b>        | Chooses CUSTOM or a procedural Factory table whose harmonic shape becomes the filter response. Choose by the broad character first, then use Frame to find the useful position inside it.                                                                                                                     |
| <b>FRAME</b>        | Chooses the base position inside the selected table and can also be dragged on the waterfall. Explore widely before refining Cutoff or Resonance, because Frame changes the complete harmonic contour.                                                                                                        |
| <b>CUTOFF (H24)</b> | Places harmonic 24 of the table at the selected frequency; it is not a conventional filter corner. Drag the response marker across the source and listen for a useful register rather than a familiar low-pass corner.                                                                                        |
| <b>RESONANCE</b>    | Changes the contrast and depth of the table-derived contour; MINIMUM, LINEAR, and ORIGINAL then apply internal automatic makeup capped at +12 dB, while RAW bypasses it. Increase contrast carefully and compare loudness externally because the internal makeup is capped at +12 dB and does not run in RAW. |
| <b>DRIVE</b>        | Adds input-dependent nonlinear color after the table filter. Its neutral position bypasses the color stage exactly. Set the filter first, then add only enough color to support the source without harsh aliasing or lost headroom.                                                                           |
| <b>MIX</b>          | Blends the latency-aligned direct and filtered paths. Use a partial blend to preserve transients when RAW or a deep contour is intentionally rough.                                                                                                                                                           |

| Control                | What it does and when to use it                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PHASE</b>           | Chooses MINIMUM, LINEAR, ORIGINAL, or RAW phase and transient character while keeping fixed alignment. Compare all four modes on the same short passage, especially on drums and at partial Mix.      |
| <b>MORPH LFO TYPE</b>  | Chooses OFF, SINE, TRI, SAW, SQR, or RND movement for the live Frame. Choose the motion shape by rhythm and continuity: smooth waves flow, SQR steps, and RND holds a new position each cycle.        |
| <b>MORPH LFO RANGE</b> | Sets a one-direction travel span above or below the base Frame; it is not a symmetric depth. Keep the base away from the table end in the chosen direction so the live Frame does not remain clamped. |
| <b>MORPH LFO RATE</b>  | Sets the free-running Frame movement speed without DAW tempo sync. Set the speed in the complete arrangement; it is free-running and will not lock to tempo.                                          |
| <b>BYPASS</b>          | Returns the fixed-latency direct signal while internal state continues for a smooth return. Compare at a matched listening level with DAW delay compensation enabled.                                 |
| <b>NEW</b>             | Creates a fresh two-keyframe Custom table. Use it when an imported or captured table is too complex to edit into the intended result.                                                                 |
| <b>CAPTURE</b>         | Copies the currently rendered frame into the matching slot of the Custom table. Capture a useful Factory frame, move to another slot, and add only enough keyframes to describe the transition.       |
| <b>DRAW FRAME</b>      | Lets you draw and normalize the waveform stored at the current Custom slot. Make broad shapes first, release to commit, and audition the interpolation before drawing finer detail.                   |
| <b>CLEAR</b>           | Removes the authored Custom keyframe at the current slot. Remove unnecessary authored frames when the transition feels crowded or changes too quickly.                                                |
| <b>TABLE FILE</b>      | Imports or exports .tbft Custom table data separately from a complete plug-in preset. Use .tbft to exchange the table only and .tbftpreset when the complete sound must be recalled.                  |

# Practical uses

## Slow pad or drone morph

- Begin with Gentle Pad Motion or Slow Vowel Bloom.
- Choose a smooth Organic Motion or Spectral Contours table and place the base Frame away from the end of travel.
- Use SINE or TRI with a slow Rate and a moderate directional Range.
- Set Mix for the desired depth and leave Drive neutral until the spectral motion is balanced.

Expected result: A sustained sound changes color and internal form gradually without requiring many moving EQ bands.

## Drum formant or stepped movement

- Start with Hollow Drum Focus or Pulse Steps and loop the clearest hits.
- Move Cutoff until the contour supports either body or attack.
- Compare MINIMUM with RAW and use SQR, SAW, or a Stepped table when a hard rhythmic change is desired.
- Reduce Mix or Resonance if the transient loses too much weight.

Expected result: Drums gain animated notches and formant character while the direct hit can remain present through Mix.

## Vocal or synth formant color

- Choose Vocal Formant Lift, Liquid Reed, or Prime Choir.
- Sweep Frame first to find the vowel or partial structure, then move Cutoff to place it around the source.
- Use ORIGINAL for more source-phase character or LINEAR when a constant-delay contour is preferred.
- Add only enough Resonance and Drive to make the formant clear without creating harsh level-dependent buildup.

Expected result: The source takes on a vocal, reed-like, or choral color that remains connected to its original performance.

## Conservative bus contour

- Begin with Subtle Bus Contour and keep Mix low.
- Use a broad, stable table region and avoid fast LFO movement.
- Set Drive to its neutral position and compare the phase modes on the complete arrangement.
- Level-match outside the plug-in before deciding whether the contour improves the bus.

Expected result: The bus receives a restrained spectral shape without turning into an obvious moving effect.

## Build and exchange a Custom table

- Open EDIT and TABLE TOOLS, then use NEW or CAPTURE to establish the first Custom frames.
- Move to another Frame slot, draw or capture a contrasting shape, and audition the interpolation before adding more frames.
- Export the table as .tbft when only the table should be shared, and save a .tbftpreset when the entire effect setting must return.

Expected result: A reusable personal filter contour can move between projects without confusing table data with the full plug-in state.

## Common pitfalls

If the sound becomes much quieter, lower Resonance or Mix and level-match externally; the plug-in has no user Output control, and the internal automatic makeup is capped at +12 dB and bypassed in RAW, so deep contours can still lose level.

Cutoff (H24) is not a conventional low-pass corner, so judge the whole response curve and the sound rather than expecting familiar cutoff behavior.

If Frame LFO motion appears to pause, move the base Frame away from the table end or reverse the sign of Range; the live position is clamped at the boundaries.

Range moves in one direction from the base Frame and does not provide a separate symmetric plus/minus mode.

The response display excludes Drive and Mix, so a visible curve cannot predict every level-dependent or partial-blend result.

High Drive on bright material may still create some aliasing, although the nonlinear-minus-linear Drive residue uses a maximum-quality true 2x polyphase-IIR oversampling path to reduce folded aliases.

RAW and partial Mix can create deliberate roughness, impulse smear, or phase coloration; choose another Phase when that character obscures the source.

The fixed processing delay requires DAW compensation and is not intended as a zero-latency live tracking path.

There is no tempo sync, user Output control, MIDI, oscillator playback, external IR import, or drag-and-drop table import in the current panel. MINIMUM, LINEAR, and ORIGINAL do include internal automatic makeup capped at +12 dB; RAW bypasses it.